Technical Data Sheet

ALCOM ABS 530/3.1 CF

(Last update: 23.04.2024)

Base Polymer	Acrylonitrile/Butadiene/Styrene/Copolymer
Filler/Additive System	8 % carbon fibres
Special Features	electrically conductive, reduced surface resistivity
Market Segment	Automotive, Machinery
Application Area	injection moulded parts
Typical Applications	housings, functional components

Pre-Drying Conditions	80 °C in a dry air (dessiccant) dryer for 2-4 h 80 °C in an air circulating dryer for 3-6 h max. moisture content <0,10 %
Processing Injection Moulding	melt temperature 230-270 °C mould temperature 60-90 °C
Storage	dry, protected from light

Properties	Value	Dimension	Test Norm
Mechanical Properties			
Flexural Modulus	6500	MPa	ISO 178
Flexural Strength	108	MPa	ISO 178
Tensile Modulus	6900	MPa	ISO 527
Tensile Strength at Break	80	MPa	ISO 527
Tensile Elongation at Break	1.2	%	ISO 527
Impact Strength (Charpy, 23°C)	18	kJ/m²	ISO 179/1eU
Impact Strength (Charpy, -40°C)	16	kJ/m²	ISO 179/1eU
Notched Impact Strength (Charpy, 23°C)	5	kJ/m²	ISO 179/1eA
Notched Impact Strength (Charpy, -40°C)	4	kJ/m²	ISO 179/1eA
Thermal Properties			
Vicat B50	102	°C	ISO 306
HDT / A (1,8 MPa)	101	°C	ISO 75-1/-2
Electrical Properties			
Surface Resistance	100	Ohm	IEC 62631-3-2
Rheological Properties			
Melt Index (MVR)	12	cm³/10min	ISO 1133
MVR temperature	220	°C	-
MVR load	10	kg	-
Shrinkage (lengthwise, 24h)	0.2 - 0.4	%	ISO 294-4
Shrinkage (lateral, 24h)	0.3 - 0.5	%	ISO 294-4

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Physical Properties

Density	1080	kg/m³	ISO 1183

Liability Exclusion
These are guide values and not a specification. The test values mentioned are representative values only and not binding minimum or maximum figures. These test values have been determined on standardised test specimens and can be affected by pigmentation, mould design and processing conditions.

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